

Binning criterion, proton-tagged angular observables and two-dimensional cross sections

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Studies of 2026-09-16/17; blind (fake data throughout)

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Abstract

Superseded in part (2026-09-26). Finding (i) is replaced by the binning review in the technical supplement (section “Binning review against the 0.68 and 0.50 migration criteria”): the conditioning threshold used here to reject the candidates is not met by most released binnings either, and with edges optimised at every bin count the review finds admissible finer binnings for $\cos\theta_\mu$, θ_μ and $\cos\theta_\pi$, and finds that the released p_μ and W_{had} fail both criteria. The rest of this note stands.

Three questions were asked of the released analysis: whether the migration-diagonal requirement could be relaxed from 0.68 to 0.50, whether the proton-tagged sample should measure the proton angle and the pion–proton opening angle, and whether double-differential cross sections are supported. This note records the answers and the machinery built for them. **(i)** Relaxing the criterion admits finer binnings in $\cos\theta_\mu$, θ_μ , p_π and the proton-tagged transverse variables, but every one of the twenty-six candidates built and unfolded in full is rejected by the conditioning of the additional-smearing matrix A_C : the criterion is not what limits this measurement, and the study is reported as a study, with no rebinning adopted. **(ii)** θ_p (four bins) and the pion–proton opening angle $\theta_{\pi p}$ (three bins) both meet the unchanged 0.68 criterion and are now extracted for FHC, RHC and the combined configuration, with generator predictions from all four generators; $\theta_{\pi p}$ is the best-conditioned new observable of the analysis ($s_{\text{min}}/s_1 = 0.30\text{--}0.48$). The selection now stores the opening angle, which required reprocessing the proton-tagged tree; the reprocess was gated on a bit-for-bit comparison of every existing branch. **(iii)** The framework supports two-dimensional measurements as flattened bins with one slice per outer bin; eight pairs were screened and are being extracted. The first results confirm what the one-dimensional study already showed: a good migration diagonal does not guarantee a usable A_C . The joint FHC+RHC background fit developed in the same period is documented separately in the control-region note.

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1 Scope, status and conventions

The signal region is blind: every “measurement” here is the per-run Poisson fake data of the released throw, and the extractions are closure tests. Three families of configuration are used and must not be confused.

- **Live:** θ_p and $\theta_{\pi p}$, in `configs/ccpilp*` with the released naming, extracted into `processed/` beside the released observables. They use binnings that meet the unchanged 0.68 criterion.
- **0.50 study:** `configs/c50/`, extracted into `processed/rebuild_c50/`. Nothing from it is adopted. The directory is deliberately outside the glob of `norm_manifest.py`, which audits the release.
- **2D study:** `configs/d2/`, extracted into `processed/rebuild_2d/`.

Screening uses the per-run exposure scales and the safe central-value weight of the extraction, and the same first/last-edge conventions as the released bin configs. Two criteria appear throughout: the column-normalised migration diagonal per true bin (the BNB criterion, > 0.68) and the conditioning of the additional-smearing matrix, $s_{\min}/s_1$ from its singular values, which the analysis note already identified as the quantity that decided $\cos \theta_\pi$ and $\theta_{\mu\pi}$.

2 The 0.50 migration-diagonal study

2.1 Screening

`macros/binning_screen_2d.C` evaluates every candidate in one pass over the processed files (the earlier `binning_screen.C` used four `TTree::Draw` calls per bin, which does not scale to a long candidate list). It reproduces the published pre-screen table exactly for FHC and RHC, which is its regression test, and it also handles two-dimensional candidates (§4). For the proton-tagged observables the maximin edge scan of `macros/tki_binning_scan.C` was generalised to take the criterion, an observable filter and the maximum bin count, and to cover W , the proton angles and the opening angle; it reproduces the two-bin transverse-kinematic-imbalance numbers of the binning README.

At 0.50 the following become admissible where 0.68 refused them (smallest diagonal, FHC / RHC): $\cos \theta_\mu$ with 6, 7 or 8 bins (0.64/0.63, 0.61/0.61, 0.60/0.59); θ_μ with 7 or 8 (0.63/0.62, 0.61/0.61); p_π with three bins and the extra edge at 0.26 GeV/c (0.51/0.53); proton-tagged δp_T with three (0.59/0.55), $\delta \alpha_T$ with three (0.52/0.50), $W_{\pi p}$ with two at 1.19 GeV/c (0.56/0.56); $\cos \theta_p$ with six and θ_p with seven. p_μ still fails (0.33/0.31 released, worse when finer), as do three-bin p_π at 0.30 GeV/c, four-bin p_n and $\delta \phi_T$ in RHC. W_{had} in two bins passes even at 0.68 (0.71/0.72) although the released configuration has six.

2.2 What the full extractions say

Twenty-six candidates (thirteen binnings $\times$ FHC and RHC) were built and unfolded with the released machinery. Table 1 gives the result. The conditioning is decisive: of the finer binnings only W_{had} (two bins) keeps an A_C with $s_{\min}/s_1$ above 0.5, and the two-bin $W_{\pi p}$ and the four-bin $\cos \theta_p$ in FHC are the only others above 0.1. Several candidates have $s_{\min}/s_1$ numerically zero to three decimals: their A_C maps every model onto a single shape, so the extraction cannot be read as a shape measurement, exactly the pathology the analysis note documented for θ_μ . Closure χ^2 values are uniformly small — which is the point: they cannot distinguish these cases, because a rank-deficient A_C makes data and prediction agree by construction.

Table 1: Candidates admitted by the 0.50 criterion, built and unfolded in full (fake data). A_C rows: range of the row sums; $s_{\min}/s_1$: conditioning of A_C ; data stat.: bin-averaged data-statistical term; unc.: range of the per-bin total uncertainty; closure: range of the per-bin ratio to the A_C -smeared truth. Rows in bold keep a usable A_C .

Candidate	Config	bins	A_C rows	$s_{\min}/s_1$	data stat.	unc. [%]	closure
$\cos \theta_\mu$ 6	FHC	6	0.40–0.52	0.110	17.0%	30–76	0.80–1.21
	RHC	6	0.09–0.44	0.085	15.8%	31–133	0.58–1.25
$\cos \theta_\mu$ 7	FHC	7	0.21–0.37	0.024	23.8%	31–143	0.51–1.30
	RHC	7	0.03–0.52	0.054	19.3%	33–121	0.71–1.17
θ_μ 7	FHC	7	0.47–0.68	0.000	11.0%	25–52	1.02–1.38
	RHC	7	0.35–0.93	0.001	10.3%	30–38	1.03–1.38
θ_μ 8	FHC	8	0.53–0.73	0.000	10.0%	25–54	0.87–1.36
	RHC	8	0.16–0.76	0.001	13.2%	28–54	0.84–1.42
p_π 3	FHC	3	0.53–0.91	0.048	14.8%	39–56	0.90–0.97
	RHC	3	0.45–0.89	0.010	12.7%	40–45	0.97–1.02
δp_T 3	FHC	3	0.37–0.89	0.001	13.8%	32–37	0.94–0.96
	RHC	3	0.32–0.86	0.000	12.0%	35–36	1.04–1.12
$\delta \alpha_T$ 3	FHC	3	0.44–0.98	0.029	10.2%	42–52	0.83–0.99
	RHC	3	0.42–0.92	0.011	9.8%	41–47	1.00–1.17
$W_{\pi p}$ 2 (1.19)	FHC	2	0.27–0.91	0.264	19.1%	41–72	0.93–0.99
	RHC	2	0.28–1.48	0.147	24.1%	47–91	0.73–1.00
W_{had} 2	FHC	2	0.80–0.90	0.579	16.3%	37–87	0.80–1.03
	RHC	2	0.76–0.86	0.541	17.1%	45–100	0.78–1.05
$\cos \theta_p$ 4	FHC	4	0.54–0.76	0.445	20.6%	34–76	0.66–1.08
	RHC	4	−0.09–0.62	0.018	36.2%	41–182	0.43–1.41
$\cos \theta_p$ 6	FHC	6	0.77–0.90	0.000	25.6%	34–84	0.54–1.06
	RHC	6	0.43–0.78	0.000	24.3%	38–72	0.75–1.48
θ_p 5	FHC	5	0.59–1.14	0.034	13.5%	33–44	0.84–1.24
	RHC	5	−0.03–0.60	0.005	24.9%	39–99	0.82–1.30
θ_p 7	FHC	7	0.61–0.98	0.000	16.4%	34–58	0.87–1.30
	RHC	7	0.32–0.87	0.000	20.7%	39–60	0.77–1.14

Conclusion. The 0.68 criterion is not the binding constraint of this analysis and lowering it to 0.50 buys nothing: the binnings it admits are rejected one level down, by the regularised response. No rebinning is proposed. Two by-products are worth keeping: W_{had} in two bins is better conditioned than the released six-bin configuration, and the criterion study produced the screening tool used for the new observables below.

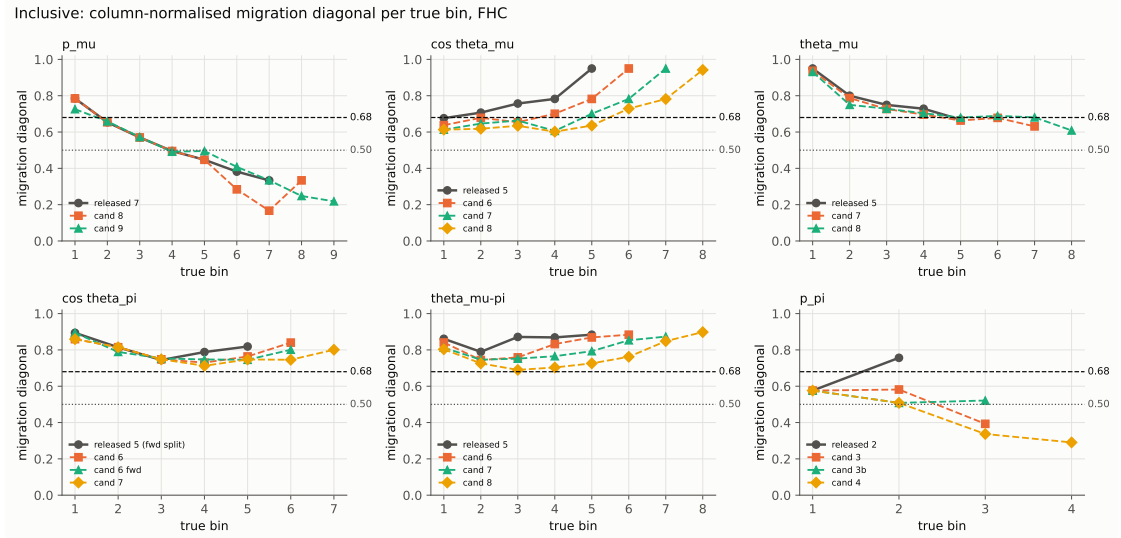


Figure 1: Migration diagonal per true bin for the released inclusive binnings and the finer candidates, FHC; the two criteria are drawn. The RHC version and the proton-tagged and two-dimensional versions are c50/diag_{incl,1p,2d}_{fhc,rhc}.

3 Proton-tagged angular observables

3.1 What was added, and why the opening angle needed a reprocess

The proton-tagged selection already stored the proton polar angle about the neutrino direction ($\cos \theta_p$, reconstructed in the beta convention, truth against the per-event neutrino). It did not store the pion–proton opening angle $\theta_{\pi p}$, nor the index of the chosen proton, so the angle could not be formed offline from the stored quantities. Both are now written by `CC1mu1pi1p`: the reconstructed angle from the two track directions (frame independent, and immune to the zero-momentum degeneracy of a scaled vector), the true angle from the leading charged pion and the leading proton.

Before committing to the reprocess, the proton choice was replicated offline (`macros/proton_angle_friend.C`) — the lowest-LLR track above threshold, contained, not the muon or pion candidate — and validated against the stored $\cos \theta_p$: identical for all 18 698 selected events of the fourteen Monte Carlo files. That friend tree also answered what the candidate is: on selected *signal* events it is the leading true proton 82% of the time, a softer proton 6%, and not a proton 13% (on all selected events, background included, 59, 10 and 31%). The wrong-candidate events populate a flat tail in both angles, while the correct ones peak within about 0.1 rad (figure `c50/resolution_proton_angles`).

Reprocess and its gate. The proton-tagged tree (41 units) and the seven per-run beam-off files, which are shared with the inclusive tree and carry both selections, were reprocessed into staging directories, and the per-run fake data re-thrown with the same seeds. Promotion was gated on `macros/compare_reprocess.C`: for every file, all 168 numeric branches of the old tree identical entry by entry, the entry count and `summed_pot` unchanged, and the new angle equal to the friend-tree value on every selected and signal event. All 56 files passed. One needed attention: the FHC Run-1 dirt snapshot carries a `summed_pot` written in place by `set_summed_pot.C`, 1.673920×10^{21} against the native 1.6739168×10^{21} ; the established value was carried into the new file so that the normalisation cache keys are unchanged. The old trees are kept as `w_pre_thpipr` and `ext_perrun_pre_thpipr`.

3.2 Binning

The maximin scan (FHC Runs 1, 2, 4; ≥ 400 events per bin) gives the largest bin count at each criterion; the weighted screen then checks it over FHC and RHC at the released exposure (Table 2). θ_p supports one bin more than $\cos \theta_p$ at both criteria, so the angle rather than its cosine is measured, matching the released θ_μ convention. Both adopted binnings meet 0.68; the RHC θ_p value sits exactly at it, as the released $\cos \theta_\mu$ does.

Table 2: Proton-tagged angular binnings. Scan: worst diagonal of the maximin scheme at that bin count (unweighted, FHC Runs 1, 2, 4). Screen: smallest diagonal FHC / RHC with the per-run weighting. Adopted binnings in bold.

Observable	bins	scan	screen FHC / RHC	edges
θ_p	4	72.6%	0.73/0.68	0, 0.597, 0.942, 1.287, π
θ_p	5	68.0%	0.67/0.64	0, 0.377, 0.691, 0.942, 1.225, π
θ_p	7	57.1%	0.56/0.55	(0.50 criterion)
$\cos \theta_p$	4	73.2%	0.73/0.66	−1, 0.275, 0.575, 0.825, 1
$\cos \theta_p$	6	57.2%	0.58/0.55	(0.50 criterion)
$\theta_{\pi p}$	3	75.8%	0.77/0.75	0, 1.193, 2.010, π
$\theta_{\pi p}$	4	67.9%	—	0, 0.973, 1.476, 2.010, π
$\theta_{\pi p}$	7	53.9%	0.52/0.53	(0.50 criterion)

3.3 Generator predictions

The four generator predictions are produced in `generator_predictions/gen2d/`, which holds copies of the released readers (GENIE `gst`, GiBUU `FinalEvents`, NuWro and NEUT in their containers) with the new observables added through one shared header, so the released chain is untouched. Two checks are built in. Every histogram the copies also write is reproduced bit-for-bit from the released per-flavour files, for all four generators; and the generic combiner reproduces all sixty released inclusive FTE histograms (four generators $\times$ FHC, RHC, combined $\times$ five observables) exactly. The new ones are produced for FHC, RHC and the combined configuration, and also supply the RHC and combined predictions of the proton-tagged W_{had} and TKI results. `scripts/check_ext_bins.py` checks that the generator edges and the configuration edges agree and that each prediction has as many bins as its configuration.

3.4 Extractions

Table 3: Extractions of the new observables (fake data). Columns as in Table 1; χ^2 is the closure against the A_C -smeared truth. θ_p is on the adopted RHC-derived edges (§3.6).

Observable	Config	bins	A_C rows	s_{min}/s_1	data stat.	unc. [%]	closure χ^2/ndf
θ_p	FHC	4	0.53–0.94	0.383	16.6%	34–64	1.92/4
	RHC	4	−0.09–0.72	0.001	21.0%	39–91	0.04/4
	COMB	4	0.74–0.99	0.408	14.7%	35–83	0.67/4
$\theta_{\pi p}$	FHC	3	0.57–0.65	0.480	20.1%	38–93	0.16/3
	RHC	3	0.59–0.73	0.376	24.5%	43–187	0.12/3
	COMB	3	0.43–0.49	0.304	14.3%	40–89	0.06/3

$\theta_{\pi p}$ is the best-conditioned observable of the proton-tagged programme: its s_{min}/s_1 of 0.30–0.48 is above everything in Table 1 except two-bin W_{had} , and above the released $\cos\theta_\pi$ (0.27/0.24). Its price is the last bin, whose total uncertainty reaches 93% (FHC) and 187% (RHC). θ_p unfolds well in FHC and the combined configuration (0.383, 0.408) but is rank-deficient in RHC (0.001), where its diagonal sat exactly at the criterion: the RHC θ_p curves should not be read as a shape measurement.

3.5 What the proton candidate is, and what that costs

Both new observables are built on the reconstructed proton candidate, while their truth-level definitions use the leading true proton. The two need not be the same particle, and the difference is absorbed by the response, so the question worth asking is not how often the pick is right but how much of the smearing it causes. Both are measured on selected signal events with the friend trees of `macros/proton_angle_friend.C` (`python3 -m bkgfit.protonmatch`, 10 105 events).

Table 4: What the reconstructed proton candidate is on selected signal events, and the reconstructed-minus-true residual of the two angles in each category. IQR is the interquartile range; the last column of each pair is the fraction of the category beyond 0.3 rad.

Candidate truth	events	share	θ_p residual [rad]		$\theta_{\pi p}$ residual [rad]	
			IQR	$ r > 0.3$	IQR	$ r > 0.3$
the leading true proton	8252	81.7%	0.073	6.1%	0.124	15.5%
a proton, not the leading one	577	5.7%	1.203	77.6%	1.439	77.3%
not a proton	1276	12.6%	0.979	74.8%	1.090	70.1%

The pick is the leading true proton 81.7% of the time and some true proton 87.4% of the time; 12.6% of candidates are not protons at all. (An earlier figure of 59% recorded during the screen is superseded: it was measured before the proton-tagged reprocess.)

The residuals say what that costs, and they say it in two different ways depending on which question is asked. A correct pick reconstructs the angle tightly — an interquartile range of 0.073 rad in θ_p , 0.124 rad in $\theta_{\pi p}$. A wrong pick does not smear the angle, it randomises it: interquartile ranges of 1.0–1.4 rad, with three quarters of those events beyond 0.3 rad. Averaged over everything the pick therefore widens the *core* by only a quarter to a third ($0.073 \rightarrow 0.109$ rad for θ_p , $0.124 \rightarrow 0.165$ for $\theta_{\pi p}$), which makes it look like a minor effect.

It is not a minor effect, because a binned measurement does not care about the core. What moves an event out of its bin is a large residual, and there the attribution reverses: residuals beyond 0.3 rad occur in 18.9% of θ_p events, and 74% **of them are wrong picks**; for $\theta_{\pi p}$, 25.9% of events, 51% of them wrong picks. The candidate purity is thus the dominant source of bin-to-bin migration in θ_p and half of it in $\theta_{\pi p}$ — which is the mechanism behind the numbers of §3: θ_p ’s migration diagonal sat exactly at the criterion and its RHC response is rank-deficient.

The control region already exists. The open item that accompanied this measurement asked for an inverted-proton-PID control region. It is already in hand: `pion_number` is defined by the same LLR cut that picks the proton (`CC1mu1piXp.cxx`), so an otherwise-signal event whose proton fails the proton PID has two pion candidates and lands in the multi- π sideband — which is open on beam-on data, and whose data-to-prediction ratio is 0.84. Two caveats keep this from closing the item. It constrains the mis-identification in the direction proton $\rightarrow$ pion, while the 12.6% “not a proton” category above is the opposite direction, pion $\rightarrow$ proton; the two are the same LLR modelling seen from two sides, not the same rate. And the multi- π region is the most signal-contaminated of the four control regions (22.1%, and 12.3% in its far variant), so it is also the one a constraint should lean on least. A dedicated region that inverts the proton PID while keeping exactly one pion candidate would separate the two directions; it does not exist yet, and building it needs no new data, only a selection flag.

3.6 Does a coarser RHC binning rescue θ_p ?

The edges above were derived on the FHC chain, because `tki_binning_scan.C` only ever chained the FHC files; it now takes a beam argument, with FHC the default so every published screen reproduces. Run on the RHC chain, the maximin scan returns different edges: three bins at 0.785 and 1.539 rad with a worst diagonal of 0.78, four bins at 0.659, 1.068 and 1.539 rad at 0.73. Both were extracted in all three configurations (Table 5), and the four-bin scheme has since been **adopted**: the live configurations, the generator predictions and the three extractions are rebuilt on it, reproducing the study builds to the last digit of σ_{int} (0.5323 , 0.3906 , 0.4846×10^{-38} cm²/Ar). Regenerating the predictions left 240 histograms byte-identical and changed only `thetap_fte`, in the twelve proton-tagged files. The superseded FHC-edge outputs are kept as `*_thetap_fhcedges`.

Table 5: θ_p on RHC-derived edges, against the released FHC-derived scheme. Columns as in Table 3.

Binning	Config	bins	A_C rows	$s_{\min}/s_1$	data stat.	unc. [%]	closure
released (FHC)	FHC	4	0.58–1.30	0.082	12.3%	35–45	0.86–0.98
	RHC	4	0.21–1.00	0.002	12.9%	36–44	0.95–1.17
	COMB	4	0.60–1.24	0.125	9.7%	32–50	0.89–1.09
RHC maximin	FHC	4	0.53–0.94	0.383	16.6%	34–64	0.80–1.06
	RHC	4	−0.09–0.72	0.001	21.0%	39–91	0.95–1.19
	COMB	4	0.74–0.99	0.408	14.7%	35–83	0.88–1.10
RHC maximin	FHC	3	0.31–0.66	0.163	11.2%	35–47	0.90–0.98
	RHC	3	0.18–0.57	0.050	26.1%	40–94	0.85–0.99
	COMB	3	0.67–0.83	0.513	11.7%	35–72	0.91–1.03

The answer to the question is no. Dropping to three bins moves RHC from 0.002 to 0.050, which is a factor of twenty-five and still far below anything usable; at four bins RHC stays rank-deficient (0.001) and one A_C row sum even goes negative. The RHC data-statistical term rises rather than falls, from 13% to 21–26%, which is what a suppressed response does to a coarser binning. θ_p cannot be quoted as an RHC shape measurement at any of the three binnings tried.

The same table says something that was not being looked for: at the *same* four bins, the RHC-derived edges are far better conditioned everywhere else — FHC $0.082 \rightarrow 0.383$ and combined $0.125 \rightarrow 0.408$ — and the three-bin scheme reaches 0.513 combined, four times the released value and on a par with the best two-dimensional pairs of §4. The released θ_p edges are therefore not the best available, and since θ_p is a new observable and nothing depended on them yet, the binning was settled before proposing it. The four-bin RHC-derived scheme was adopted: it keeps all four bins and is the most balanced across the two usable configurations, where the three-bin scheme trades FHC conditioning (0.163) for the best combined value (0.513). The price of the change is a data-statistical term that rises from 12.3% to 16.6% in FHC and from 9.7% to 14.7% combined, and a per-bin uncertainty reaching 83% rather than 50%: better conditioning here means less smearing of the truth into the measurement, not a smaller error bar.

Proton-tagged CC1 μ 1 $\pi^\pm$: new observables, CV pseudo-data (signal region blind); curves and integrated σ [10^{-38} cm²/Ar] are A_C -smeared

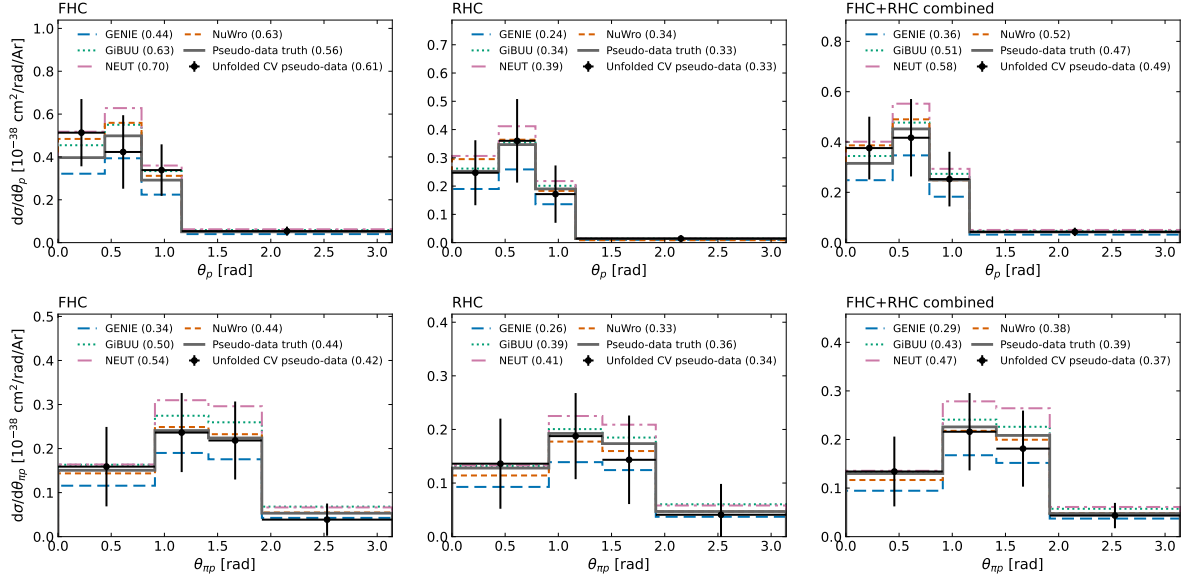


Figure 2: Extracted $d\sigma/d\theta_p$ and $d\sigma/d\theta_{\pi p}$ (fake data; the signal region is blind) against the four generators and the fake-data truth. Integrated cross sections in the legends, 10^{-38} cm²/Ar, A_C -smeared.

Both distributions are strongly forward peaked: the top θ_p bin, half the angular range, carries a fifteenth of the first bin’s $d\sigma/d\theta$ (16.6 in FHC, 14.8 combined), and the pion and proton are separated by less than 2 rad in the bulk of the events. The generator ordering is the familiar one — GENIE lowest, NEUT highest, GiBUU and NuWro between, a spread of about 1.6 in the integral — and the shapes are similar, so these observables discriminate mainly through normalisation, with NEUT’s relative excess in the middle $\theta_{\pi p}$ bin the exception.

4 Two-dimensional cross sections

4.1 What the framework already supports

A double-differential measurement needs no new unfolding machinery. UniverseMaker bins are arbitrary selection strings, so a two-dimensional binning is a flat list of bins; **SliceBinning** carries slices with an “other variable” held in a fixed range, and **UnfolderNuMI** loops over the slices and divides by the width of the other variable, so each slice is a differential cross section in the inner variable. Two constraints were found and respected: the analysis bins of a slice must be contiguous, because the per-slice A_C block is taken as a range, so the bin configuration is written slice by slice; and the closure sidecar was dumped only for slice 0, which for a two-dimensional configuration is the first outer bin rather than the whole measurement, so **UnfolderNuMI** now also dumps the “bin number” slice as `closure_hists_all_*` when a configuration has more than one physical slice (one-dimensional output is unchanged). Configurations are generated by `scripts/make_bin_configs.py`, which reproduces the hand-written one-dimensional candidate configurations byte for byte as its regression test.

4.2 Screening

Eight pairs were screened with `binning_screen_2d.C`, which for a two-dimensional candidate also splits the off-diagonal migration into leakage within the outer slice (X) and between slices (Y). Table 6 gives the best grid of each pair. The Y-leakage is what decides: the pairs built on δp_T lose 20–30% of events across the outer bin and those built on $W_{\pi p}$ up to 60%, which is why

every $W_{\pi p}$ pair fails both criteria. Expected event counts are the second constraint: $\cos \theta_\pi \times \theta_{\mu\pi}$ has a bin with four expected events.

Table 6: Two-dimensional pairs (inner $\times$ outer), best grid of each: smallest diagonal over the flattened true bins, mean leakage within and between outer slices, and the expected selected signal of the thinnest bin at data exposure.

Pair	grid	min. diagonal FHC / RHC	X-leak	Y-leak	thinnest bin
$\cos \theta_\pi \times \cos \theta_\mu$	2×3	0.69/0.68	3.4%	12.5%	25
$\cos \theta_\mu \times p_\mu$	2×2	0.70/0.68	6.1%	11.5%	69
$\cos \theta_\pi \times \theta_{\mu\pi}$	2×3	0.66/0.60	5.7%	9.1%	4
$\cos \theta_\pi \times p_\pi$	2×2	0.53/0.51	5.4%	17.2%	34
$\theta_p \times \delta p_T$	2×2	0.53/0.52	8.4%	25.4%	63
$\theta_{\pi p} \times \delta p_T$	2×2	0.56/0.55	10.7%	25.4%	58
$\theta_p \times W_{\pi p}$	2×2	0.40/0.39	8.8%	39.4%	17
$\theta_{\pi p} \times W_{\pi p}$	2×2	0.21/0.20	9.7%	39.4%	12

4.3 Extractions

All eight pairs were extracted for FHC, RHC and the combined configuration, with generator predictions produced as flattened histograms in the same bin order (Table 7).

Table 7: Two-dimensional extractions (fake data): conditioning of A_C , bin-averaged data-statistical term, range of the per-bin total uncertainty and range of the closure ratio. Pairs are ordered by their combined-configuration conditioning.

Pair	$s_{\min}/s_1$			combined configuration		
	FHC	RHC	COMB	data stat.	unc. [%]	closure
$\theta_p \times \delta p_T$	0.455	0.332	0.617	13.7%	31–88	0.82–1.25
$\cos \theta_\pi \times \cos \theta_\mu$	0.245	0.315	0.563	14.8%	32–62	0.93–1.17
$\theta_{\pi p} \times W_{\pi p}$	0.403	0.214	0.560	57.7%	43–545	−0.40–1.10
$\cos \theta_\pi \times \theta_{\mu\pi}$	0.027	0.133	0.169	22.2%	28–94	0.91–1.37
$\cos \theta_\mu \times p_\mu$	0.002	0.261	0.110	8.9%	33–57	0.97–0.99
$\theta_p \times W_{\pi p}$	0.008	0.152	0.105	111.7%	36–291	0.39–1.22
$\cos \theta_\pi \times p_\pi$	0.023	0.158	0.052	12.9%	32–58	0.96–1.14
$\theta_{\pi p} \times \delta p_T$	0.024	0.035	0.028	10.7%	28–50	0.93–1.08

Four things follow.

- **Two pairs are usable, and only combined:** $\theta_p \times \delta p_T$ (0.617) and $\cos \theta_\pi \times \cos \theta_\mu$ (0.563), both better conditioned than every released differential observable except the proton-tagged W_{had} , with data-statistical terms of 14–15% and per-bin uncertainties of 31–88%. Combining the beam modes roughly doubles their conditioning ($0.46 \rightarrow 0.62$, $0.25 \rightarrow 0.56$): this is the clearest benefit the joint exposure delivers anywhere in this work, and it comes from statistics and the shared flux, not from any $\nu/\bar{\nu}$ separation (§3).
- **The best migration diagonals give the worst conditioning.** $\cos \theta_\mu \times p_\mu$ screened best of the eight (0.70/0.68) and is rank-deficient in FHC (0.002). As in §2, the diagonal is a necessary condition, not a sufficient one, and the regularised response is what decides.
- **Everything built on $W_{\pi p}$ is unusable**, as its 39% leakage between outer bins predicted: data-statistical terms of 58–1830%, per-bin uncertainties to 5988% and closure ratios that go negative (−0.78 in RHC, −0.40 in the combined $\theta_{\pi p} \times W_{\pi p}$). This is the two-dimensional form of the conclusion already drawn for the differential $W_{\pi p}$ shape. $\theta_{\pi p} \times W_{\pi p}$ is the

sharpest case: its combined conditioning, 0.560, is as good as that of $\cos \theta_\pi \times \cos \theta_\mu$, and the extraction is still worthless. Conditioning bounds how much of the response can be inverted; it says nothing about how much data is left in the inverted directions, and the two must be read together.

- **Correlated pairs carry little.** $\theta_{\pi p} \times \delta p_T$ is poorly conditioned (0.02–0.04) where $\theta_p \times \delta p_T$ is not, because the opening angle and the transverse imbalance are built from the same hadronic system.

Figures 3 and 4 show the two usable pairs in the combined configuration, as slices in the outer variable and as maps. The slices are the form a double differential is normally published in and the one the generators are compared in; the map is the same numbers, and is included because it shows at a glance where the measurement has support. Each panel of the slice figure carries its own vertical scale: the outer bins differ by an order of magnitude, and a shared axis leaves everything but the first slice unreadable. The maps are drawn on equal-size cells labelled by their ranges rather than on the physical axes, for the same reason.

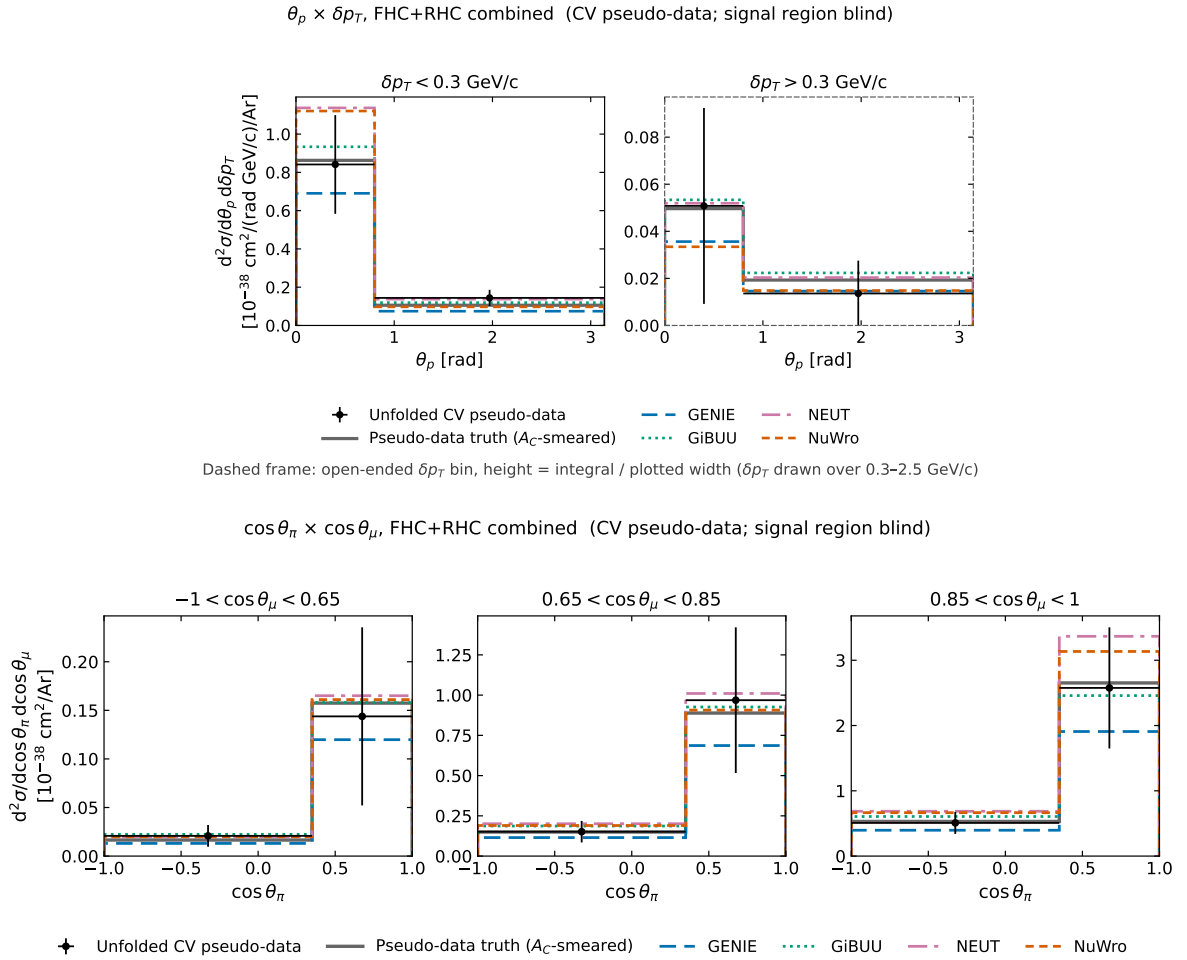
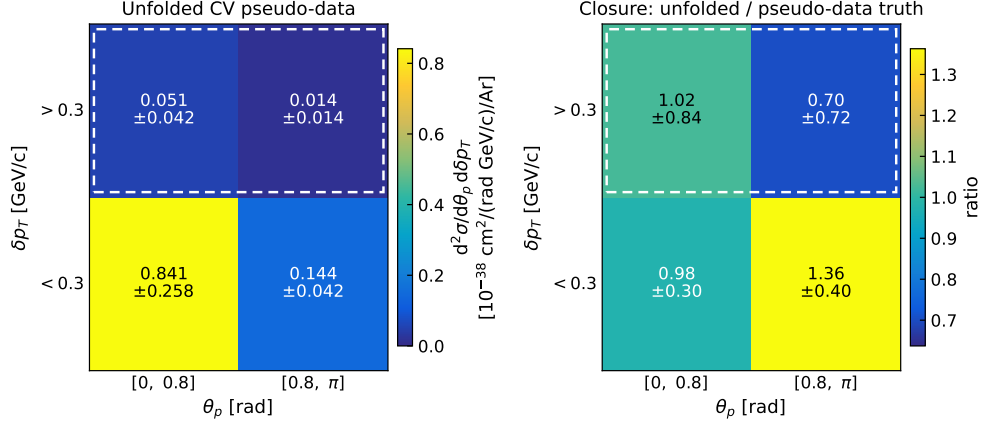


Figure 3: The two usable pairs as slices in the outer variable, combined configuration: $\theta_p \times \delta p_T$ (top) and $\cos \theta_\pi \times \cos \theta_\mu$ (bottom). Points are the unfolded **fake** data with the total uncertainty; the grey histogram is the fake-data truth and the four coloured histograms are the generator predictions, all A_C -smeared. The signal region remains blind.

$\theta_p \times \delta p_T$, FHC+RHC combined (CV pseudo-data; signal region blind)



Dashed outline: open-ended δp_T bin, height = integral / plotted width (δp_T drawn over 0.3–2.5 GeV/c)

$\cos\theta_\pi \times \cos\theta_\mu$, FHC+RHC combined (CV pseudo-data; signal region blind)

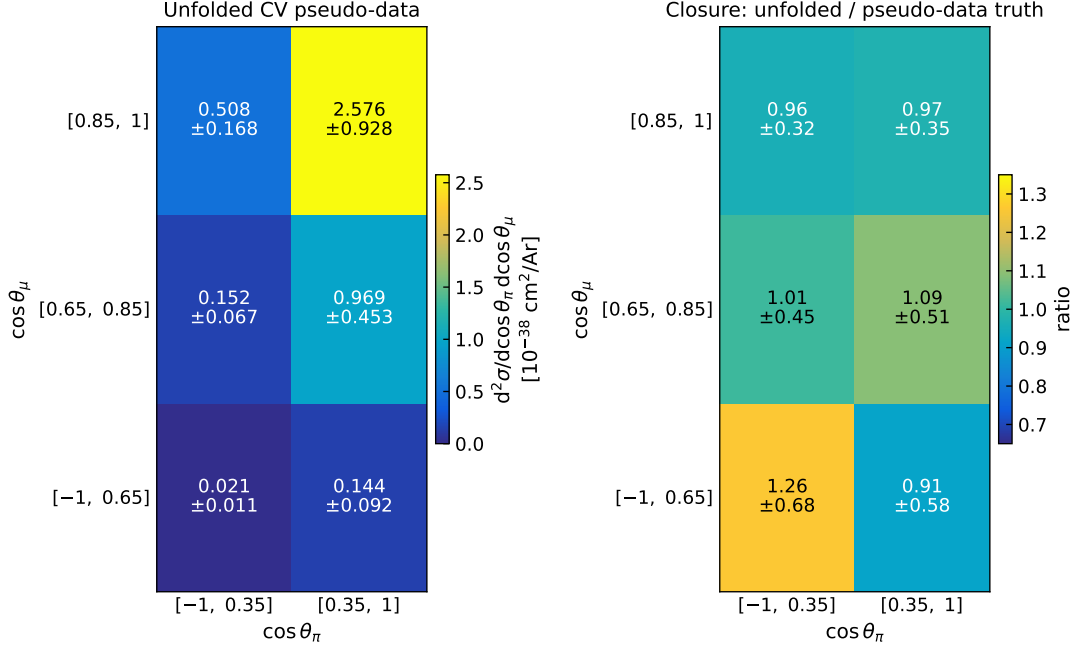


Figure 4: The same two measurements as maps: $d^2\sigma/dX dY$ per cell with its total uncertainty (left) and the ratio to the fake-data truth (right), for $\theta_p \times \delta p_T$ (top) and $\cos\theta_\pi \times \cos\theta_\mu$ (bottom). Cells are drawn at equal size and labelled by their ranges, in the **kBird** palette **UnfolderNuMI** uses for the matrix figures of the released note. The closure panel is flat to within its uncertainties in both pairs; the $\pm$ values there are the measurement uncertainty divided by the truth, so the spread of the ratios (0.82–1.25, 0.93–1.17) is well inside them.

The normalisation of the two-dimensional chain was checked against the released one-bin total extraction (`report/tools/xsec2d_total_check.py`). Summing the flattened σ over the analysis bins is not expected to reproduce the total: both are A_C -smeared with different A_C , and the row sums are not one, so each loses its own amount of normalisation — anywhere from nothing to 40% here. What must hold is that the fake data and the fake-data truth lose the *same* amount, because a wrong flux, a double-counted file or a bin area applied twice would

move the data and leave the truth. The double ratio $[\sigma_{2D}^{\text{data}}/\sigma_{2D}^{\text{truth}}]/[\sigma_{\text{tot}}^{\text{data}}/\sigma_{\text{tot}}^{\text{truth}}]$ has a median deviation from unity of 0.007 over the 24 extractions, and 20 of the 24 are within 2.5%. The four that are not (0.91, 0.94, 1.03, 1.08) are all proton-tagged RHC, the configuration whose per-bin closure is worst, and all are far inside the 38–47% uncertainty the one-bin total itself carries. The two-dimensional extraction is therefore normalised consistently with the released measurement.

Closure χ^2 values are small everywhere (0.00–0.65 on 2 degrees of freedom) and, as in §2, carry no information where A_C is rank-deficient.

5 Reproduction

```
# screening (single pass, 1D and 2D; reproduces the published pre-screen exactly)
root -l -b -q 'macros/binning_screen_2d.C+("fhc","incl")'      # rhc; incl | 1p
root -l -b -q 'macros/tki_binning_scan.C("/data/uboone/processed/w",\
    400,0.68,"thetap,thpipr",10)'
```

theta_pipr without reprocessing (friend trees, for screening only)

```
logs/c50/run_pafriend.sh
```

configurations

```
python3 scripts/make_bin_configs.py c50_candidates      # configs/c50
python3 scripts/make_bin_configs.py newobs_candidates  # configs/ + configs/d2
python3 scripts/check_ext_bins.py                      # generator vs configuration edges
# generator predictions (verifies the released histograms bit-for-bit)
generator_predictions/gen2d/regen_ext.sh
# reprocess with the opening angle, its gate, and promotion
./reprocess_thpipr.sh ; PROMOTE=1 ./reprocess_thpipr.sh
# extractions
sbatch --array=0-25%4 --export=ALL,REPO=$PWD,\
    MANIFEST=$PWD/slurm/c50_manifest.list    slurm/slurm_c50.sbatch
sbatch --array=0-29%6 --export=ALL,REPO=$PWD,\
    MANIFEST=$PWD/slurm/newobs_manifest.list slurm/slurm_extract.sbatch
```

evaluation and figures

```
python3 report/tools/binning_eval_c50.py    # 0.50 candidates
python3 report/tools/extract_eval.py        # new observables and 2D
python3 report/tools/c50_figures.py
python3 report/tools/newobs_xsec_fig.py     # Fig. 3 (1D)
python3 report/tools/xsec2d_figs.py         # Figs. 3-4 (slices, maps)
python3 report/tools/xsec2d_total_check.py  # 2D normalisation vs the one-bin total
```

6 Open items

1. Two two-dimensional pairs are proposed as released measurements: $\theta_p \times \delta p_T$ and $\cos \theta_\pi \times \cos \theta_\mu$ (Table 7, Figs. 3–4). The generator predictions are confirmed against the flattened bin order (`scripts/check_ext_bins.py`). **Decided 2026-09-18:** the combined result is the measurement, and the per-mode extractions go to Appendix A marked as not shape measurements.
2. θ_p in RHC is rank-deficient, and §3.6 shows that coarsening does not rescue it (0.002 $\rightarrow$ 0.050 at three bins): it is quoted for FHC and the combined configuration only. **Decided 2026-09-18:** θ_p adopts the four-bin RHC-derived edges 0, 0.659, 1.068, 1.539, π rad, which condition three times better in both usable configurations at the same number of bins. Configurations, generator predictions and the three extractions are rebuilt on them; the superseded outputs are kept as `*thetap_fhcedges`.
3. Two-bin W_{had} is better conditioned than the released six-bin configuration. **Decided 2026-09-18:** the six-bin scheme stays. Two bins buy conditioning by discarding almost all of the shape, and the shape is what W_{had} is measured for.

4. The proton candidate is the leading true proton 81.7% of the time on signal events (§3.5). A wrong pick randomises the angle rather than smearing it, so although it widens the core resolution by only a quarter to a third, it supplies 74% of the bin-to-bin migration in θ_p and 51% in $\theta_{\pi p}$: this is the mechanism behind θ_p 's marginal diagonal and its rank-deficient RHC response. The inverted-proton-PID control region asked for here turns out to be the multi- π sideband, already open, but it constrains proton $\rightarrow$ pion rather than the pion $\rightarrow$ proton direction that matters, and it is the most signal-contaminated of the four regions. A region inverting the proton PID at exactly one pion candidate would separate the two; it needs a selection flag, not new data.
5. The released analysis note does not yet mention θ_p or $\theta_{\pi p}$; the binning appendix and the observable tables need entries once the decisions above are taken.

A Two-dimensional results per beam mode

The two pairs proposed in §4 are measurements in the combined configuration only: in FHC and RHC separately the responses are too poorly conditioned to invert ($\theta_p \times \delta p_T$ 0.455/0.332, $\cos \theta_\pi \times \cos \theta_\mu$ 0.245/0.315, against 0.617 and 0.563 combined). Every released differential result is also quoted per beam mode, so the per-mode extractions are shown here for completeness. **They are not shape measurements and must not be read as one:** the RHC $\theta_p \times \delta p_T$ closure alone spans 0.31–1.72. What they do show is where the combined result's advantage comes from — the same bins and the same treatment, with more data.

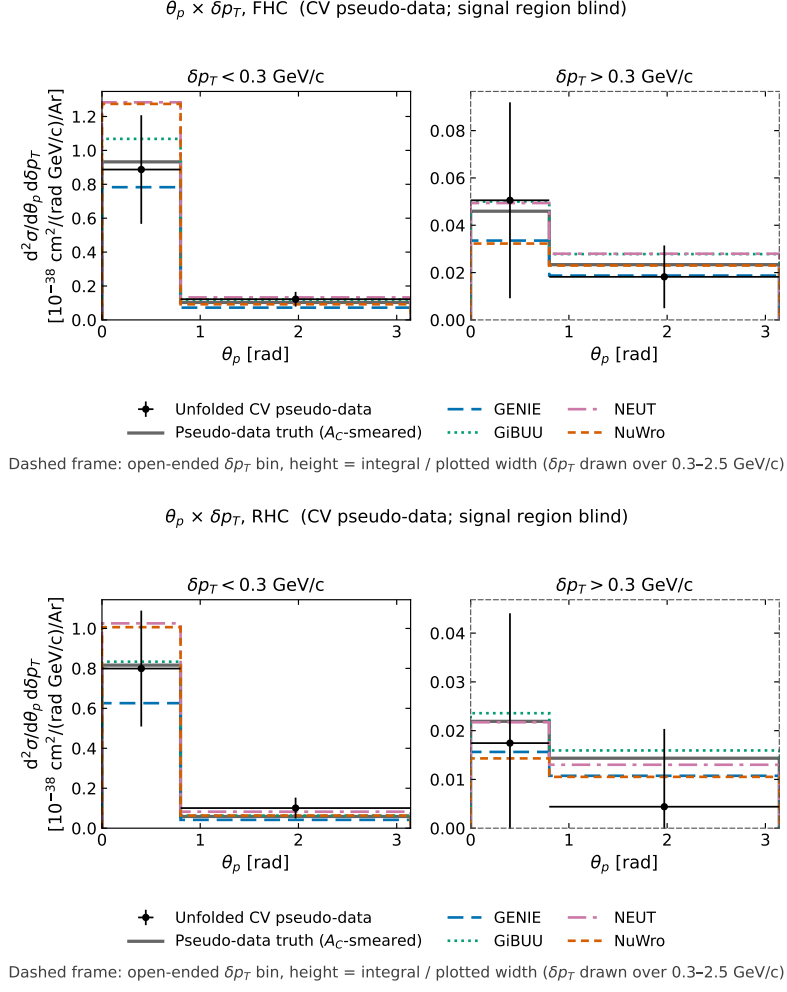


Figure 5: $\theta_p \times \delta p_T$ per beam mode, FHC (top) and RHC (bottom); conditioning 0.455 and 0.332. Curves and points as in Fig. 3.

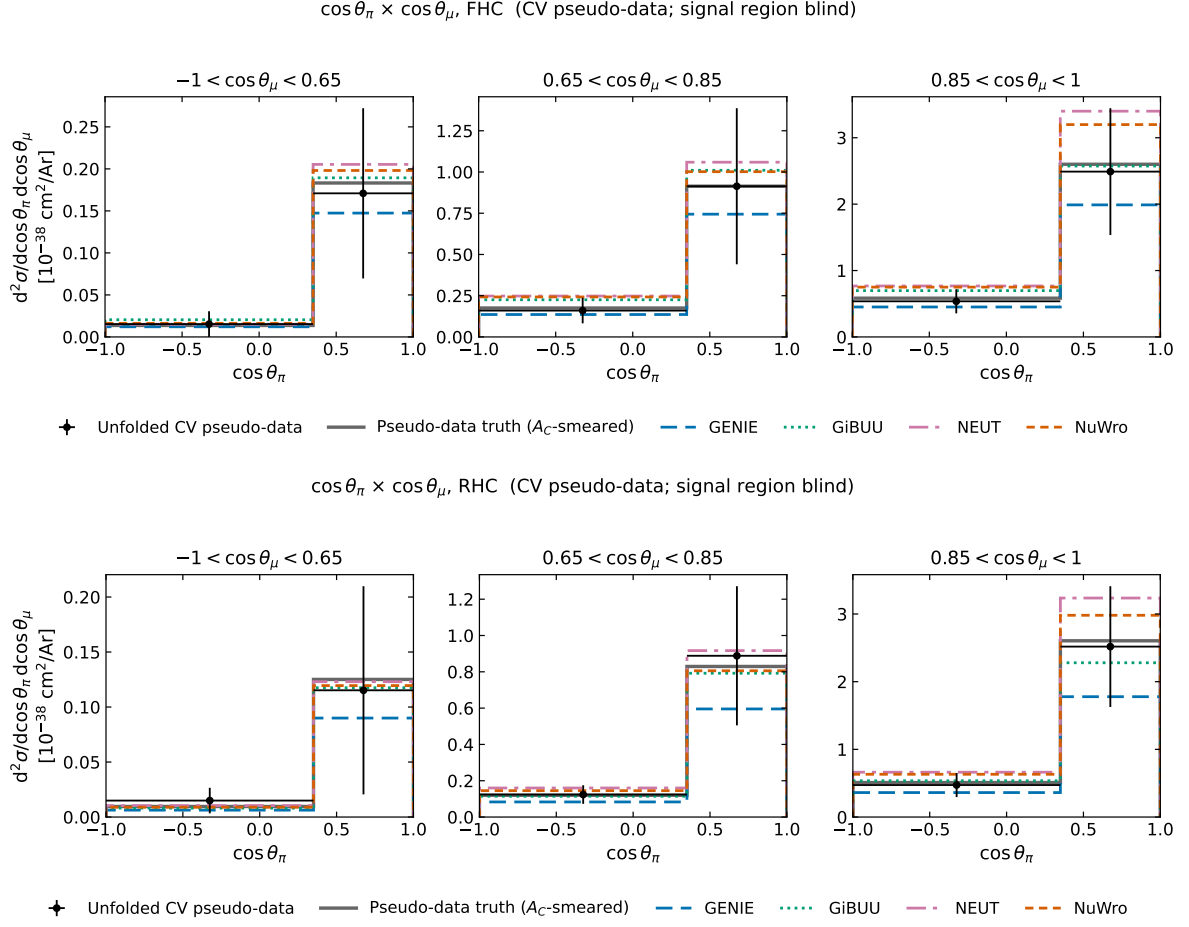


Figure 6: $\cos \theta_\pi \times \cos \theta_\mu$ per beam mode, FHC (top) and RHC (bottom); conditioning 0.245 and 0.315.